

Beetle Bot Manual

Phase 1: Workspace & Environment Setup

Before touching the robot, you need to prepare your local development environment.

1. **Create Workspace:** Open your terminal and create the directory structure:
 - `$ mkdir -p ~/ros2_ws/src` (Replace `ros2_ws` with your register number).
 2. **Initialize & Build:**
 - Navigate to the folder: `$ cd ~/ros2_ws`.
 - Source the ROS 2 Jazzy environment: `$ source /opt/ros/jazzy/setup.bash`.
 - Build the workspace: `$ colcon build`.
 3. **Automate Sourcing:** Add your workspace to your bash configuration so it loads every time you open a terminal:
 - `$ echo "source ~/ros2_ws/install/setup.bash" >> ~/.bashrc`.
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Phase 2: Turtlesim Simulation (Testing)

Use this to ensure your ROS 2 installation is working correctly.

1. **Install Turtlesim:**
 - `$ sudo apt update`.
 - `$ sudo apt install ros-jazzy-turtlesim`.
2. **Run Simulator:**
 - In a new terminal, source the environment.
 - Launch the node: `$ ros2 run turtlesim turtlesim_node`.
3. **Teleoperation (Control):**
 - Open another terminal and source it.
 - Run the keyboard controller: `$ ros2 run turtlesim turtle_teleop_key`.
 - *Alternative:* Use the general keyboard teleop with a remap: `$ ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args --remap cmd_vel:=my_cmd_vel`.

4. Topic Data view :

Ros2 topic list

Ros2 topic echo

5. Record topic data:

ros2 bag record /cmd_vel

ros2 bag record -o my_cmd_vel_bag /cmd_vel

ros2 bag record /scan

ros2 bag record -o scan_data_bag /scan

ros2 bag record -a

ros2 topic echo /scan > scan_output.txt

Phase 3: BeetleBot Hardware Connection

Now, let's bridge your PC to the actual robot hardware.

1. Network Connection:

- Turn on the bot and connect your PC to the Wi-Fi network: **BEETLEBOT_5G**.
- Enter the password: **15619xxx**.

2. SSH into Robot:

- Open a new terminal and type: `$ ssh veerobot@192.168.0.bot_No` (Example: `$ ssh veerobot@192.168.0.129`).
- Use the password: **veerobot**.

3. Set Domain ID (Critical Step):

- **On the Robot:** Source the workspace and set your unique ID:
 - `$ source ~/lyra_ws/install/setup.bash.`
 - `$ export ROS_DOMAIN_ID=129` (Use your specific robot number).
- **On your PC:** Run the exact same two commands in every new terminal you open to ensure the PC can "see" the robot.

Phase 4: Launching BeetleBot Operations

1. Start the Robot (Bringup):

- In the SSH terminal (on the robot), run:
 - `$ cd lyra_ws/.`

- `$ ros2 launch lyra_bringup robot.launch.py mode:=slam camera:=true imu:=true.`

2. Verify Topics (On PC):

- Open a new terminal on your PC and check the active data streams:
 - `$ ros2 topic list.`
 - To see specific data: `$ ros2 topic echo /scan` or `$ ros2 topic echo /imu/data_raw.`

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[!NOTE]

Technical Check: The manual mentions `$ ros2 topic node`. This is likely a typo for **`$ ros2 node list`**, which will show you all active nodes running on the network.

Phase 5: Visualization & Data Management

1. Visual Tools:

- To see the robot's sensors in 3D: `$ ros2 run rviz2 rviz2.`
- To see the camera feed: `$ rqt.`

2. Recording Data:

- Record a specific topic: `ros2 bag record -o scan_data_bag /scan.`
- Save a text log of the scan: `ros2 topic echo /scan > scan_output.txt.`

3. Save Your Map:

- Install the map server: `$ sudo apt install ros-jazzy-nav2-map-server.`
- Save the result: `ros2 run nav2_map_server map_saver_cli -f ~/my_map.`

copy files from your PC to a Raspberry Pi using SSH in the terminal, you can use the **scp (secure copy)** command.

```
scp <local_file_path> <username>@<raspberry_pi_ip>:<remote_path>
```

Example

If your Raspberry Pi username is pi, IP is 192.168.1.50, and you want to copy test.txt to the Pi home folder:

```
scp test.txt pi@192.168.1.50:/home/pi/
```

Copy a Folder

To copy a **full folder**, use -r (recursive):

```
scp -r myfolder pi@192.168.1.50:/home/pi/
```

Copy Only Files (all types) from a Path

If you want to copy **all files in a folder but not the folder itself**:

```
scp /path/to/folder/* pi@192.168.1.50:/home/pi/destination/
```

Example:

```
scp ~/documents/* pi@192.168.1.50:/home/pi/documents/
```

This copies **all file types** (.txt .py .jpg .pdf .cpp etc.).
